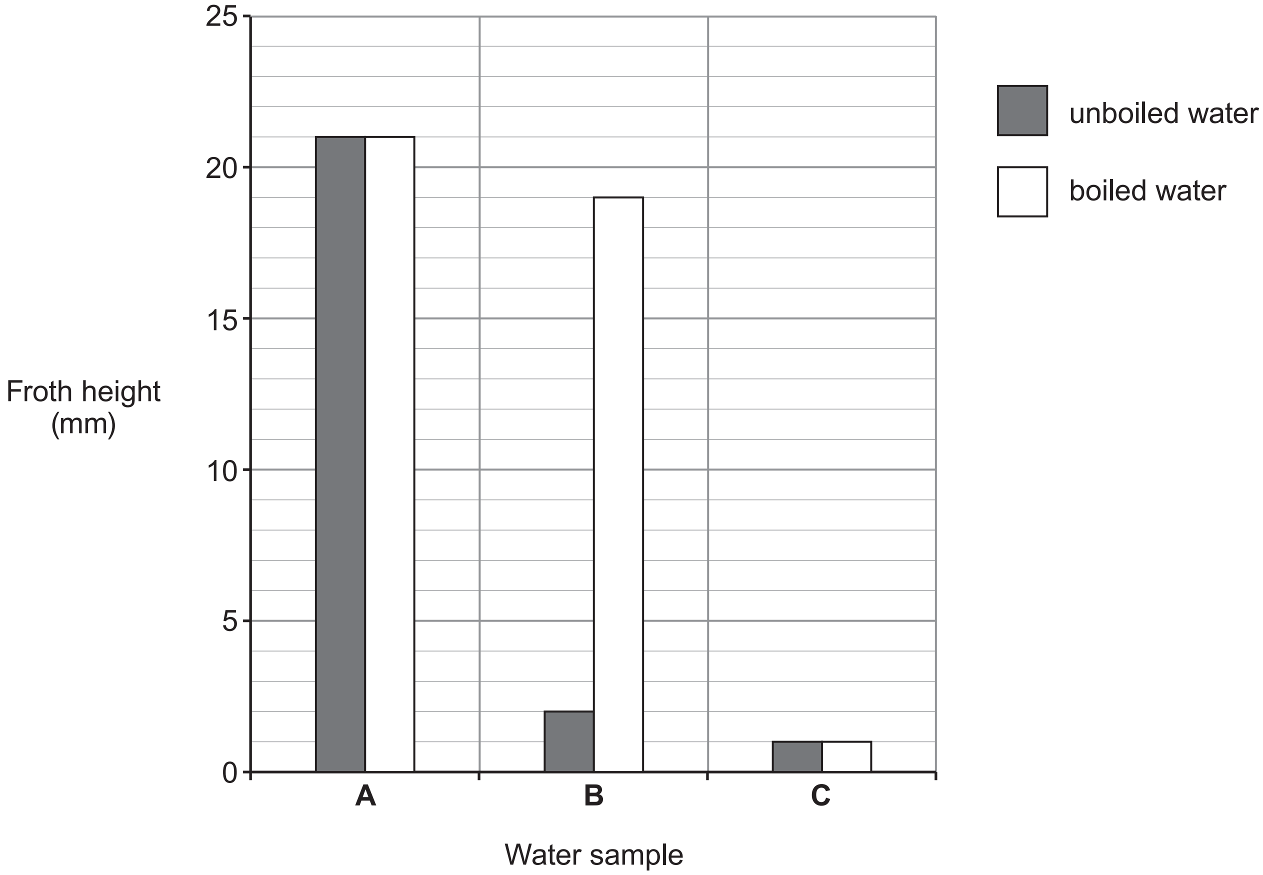
8. (a) Three samples of water, **A**, **B** and **C**, from different parts of the UK were tested in a laboratory.

$1\,cm^3$ of soap solution was added to $25\,cm^3$ of the three different water samples. Each sample was shaken for 1 minute. The height of the froth was measured.

The experiment was repeated using new samples of water, **A**, **B** and **C**, that had been boiled.



Give the **letter** of the water sample which is [2]

temporary hard water ....

permanent hard water ....

soft water ....

Examiner only

(b) There are advantages and disadvantages of living in a hard water area.

Give **two** disadvantages of living in a hard water area. [2]

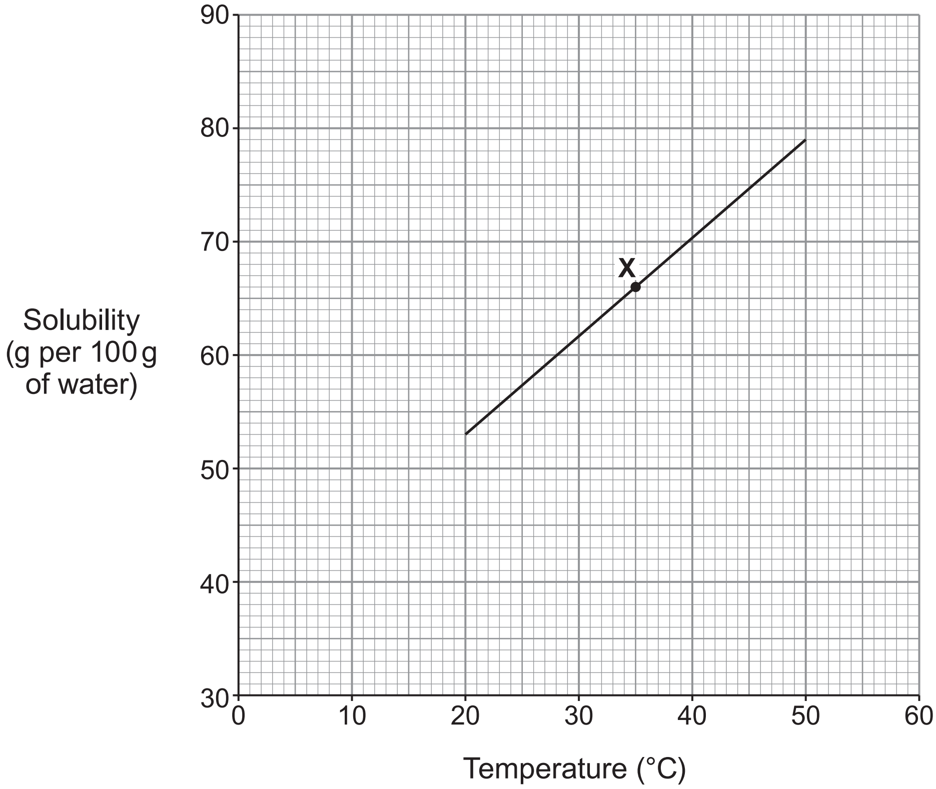
1 ....

....

2 ....

....

(c) The graph below shows the solubility of lead nitrate in water at different temperatures.



(i) State what point **X** on the graph tells you about lead nitrate. [1]

....

....

Examiner only

(ii) The solubility of lead nitrate at 20 °C is 53 g per 100 g of water.

Use the graph to find its solubility at 50 °C and hence calculate the mass of lead nitrate crystals that form when a saturated solution containing 100 g of water cools from 50 °C to 20 °C. [2]

Mass = ... g

(iii) Use the graph to find the solubility of lead nitrate at 5 °C. [1]

.... g per 100 g of water

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